Mert Torun

merttorun.github.io

Contact Information

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Profile

Electrical and Computer Engineering Ph.D. candidate and machine learning researcher with 7+ years of experience developing physics-informed deep learning, computer vision, and signal-processing systems. Published first-author work in ACM and IEEE venues; led end-to-end research in mathematical modeling, real-world applicable ML pipelines, hands-on human-subject experimentation, synthetic training-data generation, and quantitative evaluation.

Education

University of California, Santa Barbara, US — **Expected 2026**
Ph.D., Electrical and Computer Engineering | GPA: 4.00/4.00

University of California, Santa Barbara, US — **2020**
M.S., Electrical and Computer Engineering

Boğaziçi University, Turkey — **2018**
B.S., Electrical and Electronics Engineering; Minor in Economics | GPA: 3.95/4.00

- Ranked 3rd in the graduating class and 3rd across the Faculty of Engineering.

Columbia University in the City of New York, US — **2016–2017**
Study Abroad, Electrical and Computer Engineering | GPA: 3.93/4.00

Research Experience

Graduate Researcher, University of California, Santa Barbara — **2019–Present**

- **Physics-informed autoencoder for WiFi sensing:** Designed a differentiable autoencoder that maps noisy WiFi spectrograms to interpretable speed and environmental geometry; built the analytical forward model and synthetic-data pipeline used for training. Achieved 0.0715 m/s MAE on controlled motion and 0.1321 m/s across 39 walking experiments from 23 subjects, while reducing NMSE by 17.7% versus the strongest baseline.
- **Clinical gait-disorder assessment:** Co-led a one-year study with 114 participants and a survey of 70 neurologists; developed video-to-RF synthesis and machine-learning pipelines for WiFi- and camera-based gait assessment. The WiFi system achieved 85.6% per-subject gait health estimation accuracy while training primarily on synthetic RF data derived from public videos.
- **Stationary crowd counting:** Developed a statistical WiFi sensing pipeline and anomaly detector for counting seated crowds. Across 42 experiments with up to 13 people, achieved 1.04 MAE and 51 s average convergence, cutting convergence time by approximately 55% relative to the prior fidget-duty-cycle method.
- **Reflex/Danger detection:** Created a cross-modal video-to-RF modeling framework that translated public videos into robust wireless bandwidth features without collecting RF training data. Evaluated 525 natural events across four environments and 15 participants, achieving 90.15% human-reflex detection with a 2.49% false-alarm rate.
- **Wireless channel-map prediction:** Built a fully convolutional network trained on 20,000 simulated channel maps and evaluated it on 225 ray-traced environments plus real indoor measurements. At 5% spatial sampling, reduced normalized prediction error by roughly 50% versus the Gaussian-process baseline while enabling fast, size-agnostic inference.

Honors and Awards

- IEEE Internet of Things Journal Best Paper Award 2026
- Best Paper Award Runner-Up, IEEE PerCom TelMed Workshop, 2025.
- Outstanding Teaching Assistant Award, UCSB ECE Department (top three), 2019–2022.
- Bogazici and KYK Outstanding Success Scholarship (nationwide top 20), 2013
- Is Bank, Golden Youth Award (nationwide top 80), 2013
- Ranked 14th among approximately 2 million students in the Turkish University Entrance Examination, 2013.

Software Skills

Programming: Python, MATLAB, C++, C#, Git

Machine Learning: PyTorch, scikit-learn, convolutional neural networks, autoencoders, reinforcement learning

Scientific Computing: NumPy, Pandas, optimization, statistical modeling, signal processing, computer vision

Application Areas: RF/WiFi sensing, human activity and health sensing, synthetic-data generation, spatial prediction

Selected Publications

- **M. Torun**, et al., "Untangling the Geometry and Speed for RF Sensing Spectrograms," in submission process with IEEE JSAS, 2026.
- **M. Torun**, A. Parsay, and Y. Mostofi, "Fast and Robust Stationary Crowd Counting with Commodity WiFi," *IEEE Journal of Selected Areas in Sensors*, vol. 3, pp. 57–68, 2026.
- **M. Torun***, A. Parsay*, P. R. Delio, and Y. Mostofi, "Gait Disorder Assessment Based on a Large-Scale Clinical Trial: WiFi vs. Video vs. Doctor's Visual Inspection," *IEEE Internet of Things Journal*, vol. 12, no. 12, pp. 20988–21000, 2025.
- **M. Torun** and Y. Mostofi, "Wi-Flex: Reflex Detection with Commodity WiFi," *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, vol. 7, no. 3, Article 127, 2023.
- **M. Torun**, H. Cai, and Y. Mostofi, "Spatial Prediction of Channel Signal Strength Map Using Deep Fully Convolutional Neural Network," *Asilomar Conference on Systems, Signals, and Computers*, 2022.

Additional Experience

Teaching Assistant, University of California, Santa Barbara **Sep. 2018–May 2022**

Santa Barbara, CA

- Led discussion sections and office hours for Principles of Optimization, Signal Processing, Digital Design, Control Systems, and Introduction to Electrical and Computer Engineering; received the department's Outstanding Teaching Assistant Award in four consecutive years.

Software Engineer Intern, ASELSAN **Jun. 2017–Sep. 2017**

Ankara, Turkey

- Built automated test software and a user interface that checked a custom microchip for 34 known defects in 12 seconds, replacing a 15–20 minute manual process and generating structured pass/fail records for statistical analysis.

Software Engineer Intern, KoçSistem **Jun. 2016–Sep. 2016**

Istanbul, Turkey

- Developed a customer-ID-aware call-routing and auto-response system that reduced user wait time by 40% versus the non-personalized workflow.